Computer Programming Assignment 2025

MCMC sampling of the 2D Ising model

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1 Setup

We study the 2D Ising model on an $N \times N$ square lattice with periodic boundary conditions. Each spin $s_i \in \{-1, +1\}$ and the energy is

$$H(s) = -\tfrac{1}{2} \sum_i \sum_{j \in N(i)} J_{ij}\, s_i s_j, \qquad J_{ij} = 1.$$

Configurations are sampled from the Boltzmann distribution $P(s) \propto e^{-\beta H(s)}$ with a Metropolis single spin flip chain: we pick a site at random, propose $s_i \to -s_i$, and accept with probability $\min\{1, e^{-\beta \Delta H}\}$, where $\Delta H = 2 s_i \sum_{j \in N(i)} s_j$. The relevant observables are the magnetization per spin $m(s) = \frac{1}{N^2} \sum_i s_i$, the energy, and the cluster sizes.

2 Task 1: completing the code

cost(). With periodic boundaries every spin owns one right and one down bond, so summing only those two directions already covers all bonds exactly once. Using `np.roll` for the shift, the energy is just minus the sum of the right and down nearest neighbour products. This is fully vectorised and avoids any Python loop.

compute_delta_cost(). A single flip only changes the four bonds attached to the chosen site, so $\Delta H = 2 s_i (s_{up} + s_{down} + s_{left} + s_{right})$, with the modulo taking care of the boundaries. This is $O(1)$ and matches a full recomputation of the cost (checked with the `debug_delta_cost` flag of `mcmc.py`).

find_clusters(). We follow Algorithm 1: a flood fill that grows a cluster from a seed, visiting the four neighbours and keeping the ones with the same spin. We replaced the per cluster `np.random.choice` seed search (which scans the whole mask every time) with a single advancing index. The list of cluster sizes does not depend on the seed order, so this keeps the routine close to $O(N^2)$ even when there are thousands of small clusters.

3 Task 2: exploring the MCMC parameters

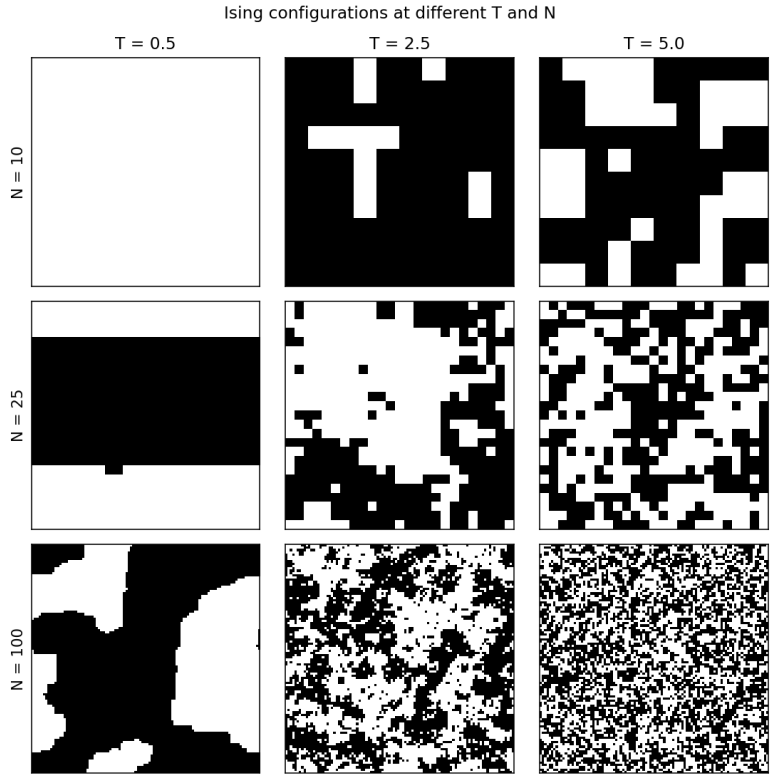


Figure 1: Configurations after burn in (positive spins in black). Columns are $T \in \{0.5, 2.5, 5.0\}$, rows are $N \in \{10, 25, 100\}$.

Figure 1 shows that aligned spins are only seen at low temperature. At $T = 0.5$ the lattice is essentially uniform (the $N = 25$ panel is caught in a long lived two domain stripe, see the burn in discussion below). At $T = 5.0$ the configuration is a random salt and pepper pattern with no order. Clusters, meaning extended domains of equal spin that are neither tiny nor system spanning, appear around $T = 2.5$, which is close to the critical temperature $T_C \simeq 2.27$. There the $N = 100$ panel shows the characteristic mix of domains of every size.

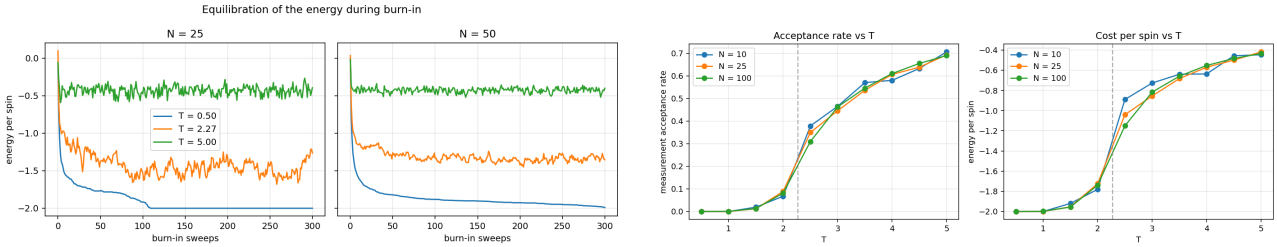


Figure 2: Left: energy per spin during burn in for three temperatures and two sizes. Right: measurement acceptance rate and energy per spin versus T for the three N. The dashed line marks T_C.

The acceptance rate (Fig. 2, right) grows with T: at low T almost every flip raises the energy and is rejected, while at high T roughly 70% of the flips are accepted. The energy per spin rises from the ground state value -2 at low T towards 0 at high T. Away from T_C both quantities depend very weakly on N, the only visible difference is that the small $N = 10$ system is noisier.

The minimal burn in needed for a roughly constant cost clearly depends on both T and N. At high T the energy reaches its plateau in a handful of sweeps. At low T it takes longer because the system has to grow a single domain, and for the larger lattice it relaxes more slowly. The hardest case is near T_C: the energy keeps fluctuating with large amplitude and never settles to a sharp value, which is the well known critical slowing down. In practice we used a burn in of the order of one hundred to a few hundred lattice sweeps (one sweep equals N^2 proposed flips), with the longer values reserved for temperatures near T_C and for the larger lattices.

4 Task 3: statistical properties and critical temperature

For this part we fixed $N = 32$, used a cold (ordered) start to avoid the striped metastable states that trap the chain below T_C, and pooled 3 independent chains with a longer wait near T_C (where samples are strongly correlated). We scanned β from 0.1 to 1.0 with a finer spacing around β_c.

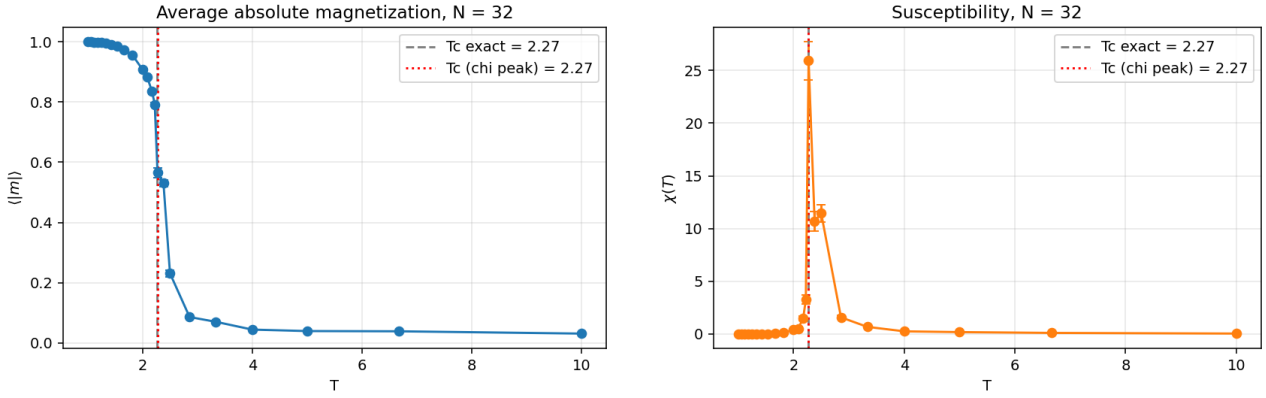


Figure 3: Average absolute magnetization and finite size susceptibility versus T for $N = 32$. The exact $T_C = 2.269$ and the susceptibility peak are marked.

Critical temperature. The magnetization $\langle|m|\rangle$ (Fig. 3, left) stays close to 1 in the ordered phase and drops quickly around $T \simeq 2.3$. Reading T_C as the highest T where $\langle|m|\rangle$ is still clearly above its finite size floor gives about 2.5. The susceptibility (right) is much sharper: it has a clear peak at $T_C \simeq 2.27$, in excellent agreement with the exact value 2.269. We adopt the susceptibility estimate as our value of T_C, since on a finite lattice the peak position is a cleaner estimator than the magnetization onset.

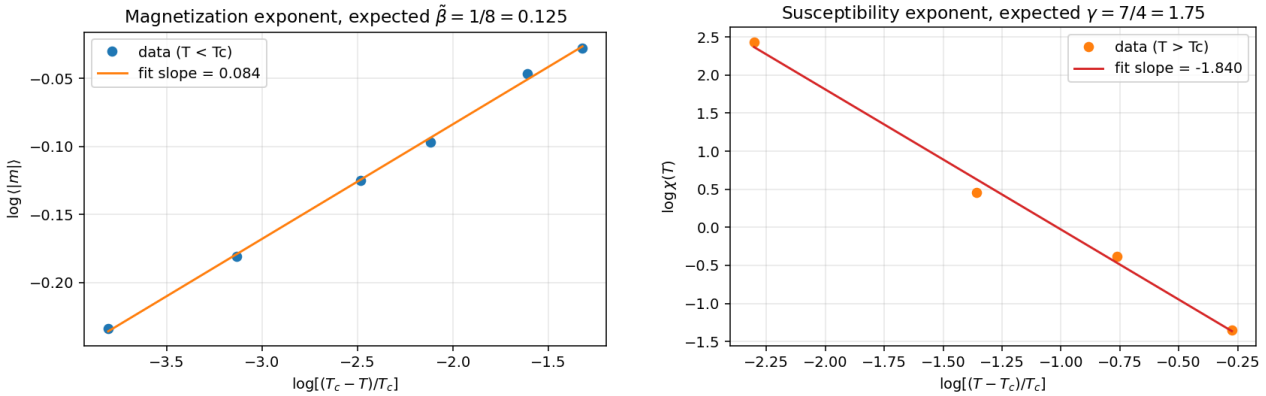


Figure 4: Power law fits. Left: $\log\langle|m|\rangle$ versus $\log[(T_C - T)/T_C]$ for $T < T_C$. Right: $\log \chi$ versus $\log[(T - T_C)/T_C]$ for $T > T_C$.

Critical exponents. Fitting the magnetization below T_C (Fig. 4, left) gives a slope $\tilde{\beta} \simeq 0.084$, to be compared with the exact $1/8 = 0.125$. The susceptibility above T_C (right) gives $\gamma \simeq 1.84$, close to the exact $7/4 = 1.75$. The susceptibility exponent comes out well, while $\tilde{\beta}$ is underestimated. This is expected: on a small lattice the magnetization is rounded near T_C, and the fit is very sensitive to the value of T_C used (a slightly higher effective critical temperature would push the slope towards $1/8$). We kept the same T_C for both fits to stay consistent. A larger lattice and longer runs would tighten both estimates.

Cluster sizes. Figure 5 shows the cluster size distribution at the three temperatures for $N \in \{10, 50, 100, 200\}$, on log log axes. At $T = 0.5$ there is a single cluster covering the whole lattice (one point at size N^2). At $T = 5.0$ there are many small clusters and the distribution falls off fast, with a cutoff at small sizes. Around T_C the picture changes completely: the distribution becomes broad and roughly straight on log log axes, that is approximately scale free, and a single spanning cluster appears (for $N = 200$ it already reaches about half of the lattice). This is the cluster signature of the transition: domains of all sizes coexist and the system is on the verge of percolating, which is exactly what makes T_C special.

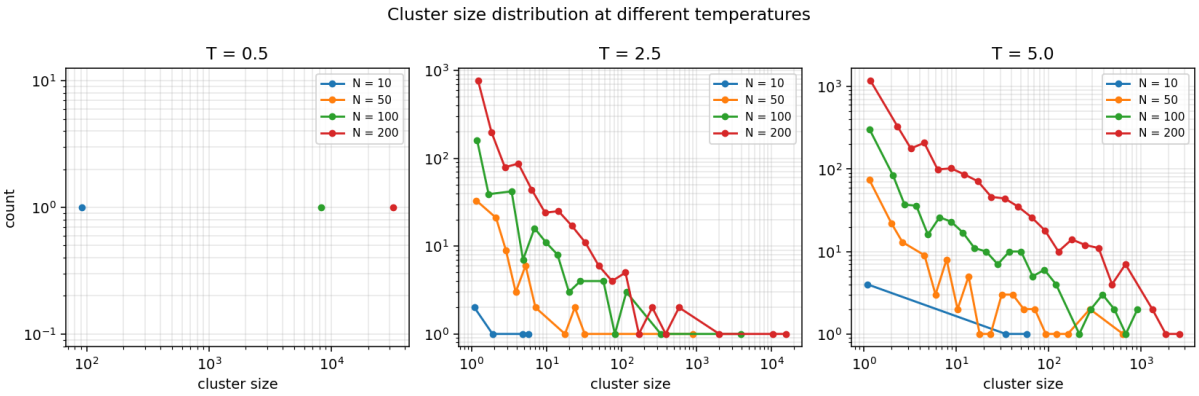


Figure 5: Cluster size distribution at $T \in \{0.5, 2.5, 5.0\}$ for several N (log log axes).

5 Conclusions

The single spin flip Metropolis sampler reproduces the two phases of the 2D Ising model and locates the critical temperature accurately from the susceptibility peak ($T_C \simeq 2.27$ against the exact 2.269). The critical exponents come out as $\gamma \simeq 1.84$ (exact 1.75) and $\tilde{\beta} \simeq 0.08$ (exact 0.125), the magnetization exponent being the harder one to measure on a finite lattice. The choice of parameters matters a lot: a cold start removes metastable stripes below T_C, and a long wait near T_C is needed because of critical slowing down. The cluster analysis gives a clean and intuitive view of the transition, from one giant domain at low T, to a broad scale free distribution at T_C, to many small islands at high T.